

MH



पश्चिम रेलवे
Western Railway

Loco Care Centre Sabarmati

IOH MAINTENANCE SCHEDULE FORM SECTION - EMH

(Loco Type: WAG 9HC Locomotives)

Ref: Standard of Inspection as per RDSO.L.No:EL/3.6.1 Date:

Loco No: 32461 Class WAG 9H Date of Sch. Started 12-1-26 Date of Sch. Completed 24/03/26
Name of L/E/SSE Dinesh Patel

	General Inspection	Final Inspection
Date/Shift	12-01-2026, 8 to 17	24/03/26 - 8/17
Name of Technician	copied to	passed.
Designation	MCF	
Signature		

Customer Feed Back/Bookings/Fault archive (Record feedback booked by Loco Pilot)			
S No.	Customer Feed Back/Bookings/Error log data	Action Taken	Name of TCN/Attended By
		MIL	

Non schedule work noticed during Incoming Inspection			
S No.	Details of Non schedule work	Action Taken	Name of TCN/Attended By

Repeated Non schedule work (As advised by MI Cell/Data download cell)			
S No.	Details of Non schedule work	Action Taken	Name of TCN/Attended By

Signature of JI



Loco Care Centre Sabarmati
IOH MAINTENANCE SCHEDULE FORM SECTION - EMH
(Loco Type: WAG 9HC Locomotives)
Ref: Standard of Inspection as per RDSO.L.No:EL/3.6.1 Date:

Inspection Wing Mechanical Heavy: Check and record following during Incoming Inspection -

Loco no. 30461 Schedule: IOH Date of Arrival: 12-1-26 Date of release: _____
Name of Supervisor: Dinesh Patel

BG-1 T/F Oil Pump Temp No 1 (In °C)	BG-2 T/F Oil Pump Temp No1 (In °C)	BG-1 T/F Oil Pump Pressure No 1 (In bar)	BG-2 T/F Oil Pump Pressure-No 1 (In bar)		BG-1 T/F Oil Pump Temp No 2 (In °C)	BG-2 T/F Oil Pump Temp No 2 (In °C)	BG-1 T/F Oil Pump Pressure No 2 (In bar)	BG-2 T/F Oil Pump Pressure-No 2 (In bar)
Std	Std	Std	Std		Std	Std	Std	Std
25.2	25.9	1.5	1.4		25.9	25.9	1.5	1.4

Transformer Pump -01-analog Gauge Pressure- 2.0 Kg/cm²

Transformer Pump -02- Analog Gauge Pressure- 2.0 Kg/cm²

VCB Pressure regulator setting- 4.51 kg/cm²

Signature and Name of Technician for Inspection Incoming with date & shift: Shrikant - 12-1-26 - 8 to 13

Signature and Name of JE/SSE for Inspection Incoming with date & shift: Dinesh Patel - 12-1-26 - 5 to 13

SN	DESCRIPTION	INCOMING	OUTCOMING
1.	Main Transformer Sr. No.	<u>65-04-19-002033</u>	
2	Type	<u>6531</u>	
3	Model	<u>04-19</u>	
4	Manufacturing Year/date	<u>04-19</u>	
5	Age on date	<u>077 can</u>	
6	Transformer received from and date	<u>CLC</u>	
7	Remarks		

A	Incoming Inspection				
SN	Details of Activity	Standard Value	Observed Value	Name of TCN	Name of JE/SSE
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

Shrikant
Signature of JT

SN	Works to be carried out	Tools Required	Standard	Actual	Sign of TCN
A	Roof Body:				
I	All hood No. 1,2 & 3 to removed.	Removed		Yes	
II	All hoods to fit after completion of schedule work. Check the tightness of roof bolts and roof joint support clamps and sealing to do.	Fitted. Done.		after fit & seal done Sealing done.	
III	Roof hatch rubber seals; replace the rubber seals around the pantograph and converter roof hatches, and the roof beam.	Completed.		Completed & done.	
IV	Check the various louver panels to ensure they are clean and free from debris. To remove dirt. Wash vents with detergent/water solution and thoroughly clean with a stiff non-metallic brush.	Checked ok		Yes	
V	Roof line DPT to do.	Carry out		Yes & done	
B	Main Transformer Sr.NO. 65-04-19-00-2073-2019 Make: HAL DOC: 2019				
I	Perform the sample test on transformer oil. Check specific value of BDV and moisture, acidity & DGA as per- RDSO/SMI/138	Pipe wrench, Special Tool	as per- RDSO/SMI/138 >60 KV DGA as per SMI 138	57.90 ...KV (After filtration)	
II	Oil centrifuging to be done in TOH/IOH as per RDSO SMI 158.	Oil filtration plant 3000 LPH	24/72 Hrs.	Completed	
III	Check the torque value of TFP core bolt on the with bright steel bars. As per RDSO/SMI NO.123.	Pipe wrench, Special Tool	25 kgm/180 ft/lb	Done	
IV	Transformer pipe line oil is to be drained for T-pump changing work in TOH/IOH schedule.	Spanners	Drained	Done	
VI	Both Transformer pump to remove in IOH sch. and send to section (EA) for O/H work.	(RDSO/TC/0 76)	Remove	Done	
VII	Replace the 'O' rings in the main transformer and main convertor oil pipes.			Changed	
VIII	BOTH TRANSFORMER PUMP TO FIT AND INLET AND OUT LET CONECTION TO DO.	(RDSO/TC/0 76)	fitted	Fit	
IX	Replace the 'O' rings in the main transformer and main convertor oil pipes.			Change done	
X	Visually inspect the condition of oil cooling metallic pipes, check for leakage / damage & check all fixing clamps and also check the drain cock, isolating cock for oil leakage & clean it.		Checked / Intact / No leakage / cleaned	Check Metallic clean	

Signature of JE/SSI

XI	TRANSFORMER OIL TO FILL AND FILTRATION TO DO BDV TO CHECK. As per RDSO SMI 158.	Oil filtration plant 3000 LPH	24/72 Hrs.	Carry out	
XII	Visually inspect the insulator and condition of insulator for any crack or damage.	-	Intact	Check. Insulator	
XIII	TRANSFORMER ALL BUSHING TO CHANGE IN IOH SCH. AND CLEANING TO DO. Replace the all terminals gaskets & "O" rings in IOH sch. of loco. (RDSO/TC/076)	(RDSO/TC/076)	Changed	Change. Insulator	
XIV	TRANSFORMER BHUSING WINDING IR VALUE TO CHECK.		check	Carry out	
XV	IR Value	2.5 KV Meggar, Multimeter	100MΩ	If value less dry out the transformer	Carry out
	Earth to GOD winding- (XB11, XB12, XB13, XB14, XB21, XB22, XB23, XB24, XB31, XB32, XB33, XB34)				
	Earth to Reactor winding- (X21, X22, X11, X12)				
	Earth to Traction winding – (2U2, 2V2, 2U3, 2V3, 2U1, 2V1, 2U4, 2V4)				
	Earth to Hotel load winding – (2UH1, 2VH1, 2UH2, 2VH2)				
	Earth to Filter winding – (2UF, 2VF)				
	Earth to Primary winding – (1V, 1U)				
	Continuity of 1U to 1V 8.47 mΩ				
XVI	Traction winding connections of transformer terminals during IOH schedule is to be done. As per RDSO/SMI/228	Torque wrench 150 Nm	Done	Done	
XVII	Check the main TFP and its protection cover for damage/crack & oil leakage. (RDSO/TC/076)	-	Checked & Found intact	Check OK, Intact	
XVIII	Check conservator foundation welding and conservator stand bolts for tightness. Check the conservator tank for any deformation and crake. AS PER SMI NO.102.	AS PER SMI NO.102.-	Intact and no craked	Carry out	
XIX	Replace Cork sheet gaskets of conservator tank bottom plate in IOH schedule.	-	Intact/ Replaced	Replaced	
XX	Oil to fill in conservative tank. Read off the oil level on the gauge situated on the conservator. Top up the oil as necessary and check for any signs of leakage.	-	Middle strip +/- 6" No leakage	No. leakage	
C	Vacuum Circuit Breaker (VCB) Sr, no: 1961416 Make: AAL. DOC: XXXX		Rem	Carry out	
	Remove the VCB and Earthing switch. from loco.		Rem		
I	Overhaul the VCB in TOH/IOH/POH as per OEM's Instructions & replace the items as per RDSO's SMI No ELRS/SMI/0236(REV.1) and OEM's instructions. Travel characteristics of VCB to check as per maintenance manual of respective manufacturer.		OH	Unit	Carry out
II	Check the healthiness of Vacuum Switch Tube (VST) (Dielectric test 40 kv for 10 sec)		OH	"	

Signature of JE/SSI

III	Check the closing & opening speed of VCT, First after 3 years on new breaker & then every year.			Check.	Calibration.
IV	Parameter	Standard m/s	Actual	{ checked OK }	
	Closing Speed	0.8 - 1.2 m/s	1.0 m/s		
	Opening Speed	1.16 - 1.75	1.54		
V	Measurements of main contact wear of Vacuum Switch Tube (VST) as per respective maintenance manual, first after three years on new breaker and then every year. Change the VST assembly with fully worn out contacts.			Check	Calibration.
VI	Check the setting of pressure switch, on test bench. As per RDSO /SMI/0208-99(Rev-0)			Close at 4.6 kg/cm ² Open at 4.0 kg/cm ²	4.6 4.0
VII	Check closing time, opening time, EV on time & counter operation Electronic Control Unit.			Checked OK	
VIII	Do maintenance of Air dryer (Refer ELRS/SMI/137 dated 01.02.1991)			Check OK	
IX	Lubricate toggle mechanism & airflow regulator rod.			Check OK	
X	Ensure proper VCB base gasket on roof to avoid entry of rain water as per respective maintenance manual of respective manufacturer.			sealant	sealing Seal
XI	Ensure settings of Pressure Regulator Valve using a master pr. gauge. Fitted in BTIL make VCBs as per RDSO's letter No. EL/3.2.61/BT dated 23.05.11				5.0 kg/cm ² 5.0
XII	O/H VCB to fit & sealing to do.			Fit & Seal	Calibration.
D	Main Circuit Breaker Earthing Switch				
I	Earthing switch (HOM) to remove.			Remove	Calibration
II	Overhaul the Earthing switch.(HOM)			OK	Calibration
II	Check that when overheating the arm of VCB Earthing switch, the contacts make proper contacts on both sides, check that copper braided is undamaged and properly connected.			Check	
III	O/H Earthing Switch to fit & sealing to do.			Check	
IV	HV hushing rubber gasket to change.			Change	
E	Traction Converter (Propulsion System) SR1 and SR2				
I	Expansion Tank SR 1 AND SR 2 COOLENT TO DRAIN AND CON.TO REMOVED.			Remove	Calibration
	BOTH SIDE SR PUMP AND RADIATORE INLET AND OUTLET PIPE GASKET TO CHANGE.WITH NEW GASKET.			Change	Calibration.
II	PH value and Ten delta test of SR oil and topping up of oil if required.			—	
III	Examine the two flange joints for leaks, loose or missing screws. Check gaskets/oil seals. Dismantle, if required. Check that the radiator is not blocked.			Check	
IV	SR1 and SR 2 coolant to fill			Fit	Calibration.

Signature of JE/SSI

F	Machine Room Blower Filter (MRB)				
I	Remove and clean the machine room blower filter duct with air and then clean with water. Remove all traces of dirt, dust and debris and refitted. As per RDSO/SMI NO.0287	Cam			Yashwanth
II	Check the condition of hoses and clamps. Replace if required.	Checked			
III	Clean the filter panel with hot water jet with suitable soap/solvent (Temp.65 °C to 70 °C) As per RDSO/SMI/0286		(Temp.65 °C to 70 °C)		Yashwanth
IV	Remove the fly screen and louvers and clean them. Rectify any fault found.	Clear			Yashwanth
V	Remove any buildup of the dirt and foreign material from around the louvers and filter panel. Also remove any obstruction from the louvers panel.	Clear	Temp. of water 65-70 deg. C		Yashwanth
VI	Filter duct wear plate of MRB should be checked and adjusted, if required.	Clear	Air pr. 3-4 kg/cm ² from 3.0 ft away		Yashwanth
VII	Replace the seal between blower and filter duct of MRB	Tools Required	Standard	Actual	Sign of TCN
VIII	Check the condition of rubber hose pipe and clamps. Replace if damaged. Check the tightness of clamps.	Carry out			Yashwanth
IX	Check the condition of the filter screen, inspect for cracks or damage. Rectify any faults found.		Checked		"
X	Check the security of the bolts around the duct flanges. Check all flanges on the ducting for cracks around the hoses. Rectify any fault found.		Cam		Yashwanth
XI	Remove any buildup of dirt and foreign material from around the filter screen.		Cam		Yashwanth
XII	Check the seal between the filter screen and pantograph roof hatch. If necessary, conduct a water test Replace the seal if any leakage occurs.			"	"
XIII	Check the seal between the filter screen and pantograph roof hatch. If necessary, conduct a water test Replace the seal if any leakage occurs.			"	Yashwanth
XIV	Replace the seal between blower and filter duct of OCB As per RDSO/SMI 0286			"	Yashwanth
G	Oil cooling unit radiator				
I	Dust particles accumulated on top side of radiator to be cleaned as per RDSO/SMI/0287		Air pr. 5-6 kg/cm ²		Yashwanth
II	Remove all dust, dirt and debris from the radiator chamber via the machine room access cover using vacuum cleaner				Yashwanth
III	Clean radiator by high-pressure hot water jet. Water tem. 65-70 deg. Pr. 5-6 kg/cm ² as per RDSO /SMI 0287		Clear		"
IV	Examine flange joint for sign of cracks, oil leakage and loose/missing screws		"		"
V	Ensure tightness of radiator protection mesh fasteners		Form		Yashwanth
VI	Check oil cooler radiator for any oil leakage/external damage from top and bottom.		Clear		"
VII	Check leakage from Transformer and traction converter "O" ring flanges of cooling pipe joints.		Noted		Yashwanth
VIII	Check tightness of foundation bolts oil pumps. (Transformer & traction converter)		Form		"
H	Machine Room Blower Filter (MRB)				

Signature of JE/SSI

I	Remove and clean the machine room blower filter duct with air and then clean with water. Remove all traces of dirt, dust and debris and refitted. As per RDSO/SMI NO.0287		cleaned	Ynt.
II	Check the condition of hoses and clamps. Replace if required.			Am Ynt.
III	Clean the filter panel with hot water jet with suitable soap/solvent (Temp.65°C to 70°C As per RDSO/SMI/0286		Cloet	
IV	Remove the fly screen and louvers and clean them. Rectify any fault found.		Cloet	
V	Remove any buildup of the dirt and foreign material from around the louvers and filter panel. Also remove any obstruction from the louvers panel.		Cloet	
VI	Filter duct wear plate of MRB should be checked and adjusted, if required.		cleo	
VII	Replace the seal between blower and filter duct of MRB		Am	
I	Fire Extinguishers			
I	Check the condition of CO2 gas cylinder, its regulator and pressure gauge in both cabs. Check the CO2 gas pressures. Refill if necessary. Replace hose pipe if required.		- Check	reg. ch. 1.1.
II	Clean pipeline by compressed air and check that there should not be any leakage.		Check Ho	Ur.
III	CO2 Gas pressure and date: Cab-1, Cab-2 - 6.4/2012		Clin	
IV	Clean fire detection unit pipeline air intake holes with soft non-metallic brush and compressed air.		Am	Collection
V	Clean pipe line by compressed air and check that there should not be any leakage.		Am	Power
J	Traction Motor Blower Filter (TMB)			
I	Remove and clean the traction motor blower filter duct with air and then clean with water. Remove all traces of dirt, dust and debris and refitted. As per RDSO/SMI NO.0287		Cloet	
II	Clean the filter panel with hot water jet with suitable soap/solvent (Temp.65°C to 70°C As per RDSO/SMI/0286		Cloet	Ynt.
III	Filter duct wear plate of TMB should be checked and adjusted, if required.		Cloet	
IV	Replace the seal between blower and filter duct of TMB As per RDSO/SMI 0286		Am	
K	Air dryer			
I	Air dryer weight to check. If found 0.8 kg more than actual weight (3.9) than change the sieve.		3.9 kg	3.6 kg
II	Shower test to do.		Cloet	Ynt.

Inspection Wing Mechanical Heavy: Check and record following during Final Inspection-

BG-1 T/F Oil Pump Temp No 1 (In °C)	BG-2 T/F Oil Pump Temp No 1 (In °C)	BG-1 T/F Oil Pump Pressure No 1 (In bar)	BG-2 T/F Oil Pump Pressure-No 1 (In bar)	BG-1 T/F Oil Pump Temp No 2 (In °C)	BG-2 T/F Oil Pump Temp No 2 (In ° C)	BG-1 T/F Oil Pump Pressure No 2 (In bar)	BG-2 T/F Oil Pump Pressure-No 2 (In bar)
38.9	39.8	39.7	39.7	39.7	39.5	1.4	1.4
Transformer Pump -01-analog Gauge Pressure-				Kg/cm ²			

Signature of JE/SSI

Transformer Pump -02- analogy Gauge Pressure-	1.4 Kg/cm ²
VCB Pressure regulator setting-	5.8 kg/cm ²

Signature and Name of Technician for Inspection Final with date & shift: Deepak Sharma - 21/1/19

Signature and Name of JE/SSE for Inspection Final with date & shift: Deepak Sharma - 21/1/19

Loco No: 32461 Class WAG9H Date of Sch. Started 12/01/26 Date of Sch. Completed 24/03/26
 Name of JE/SSE _____

Record all Modifications carried out in the following table [If required add extra sheet(s)]				
S No.	Date	Details of Modifications	Name of TCN	Sup's Sign
	Shift			

Record all Non-schedule work in the following table [If required add extra sheet(s)]					
S No.	Date	Work Done	Name of TCN	Work Balance	Sup's Sign
	Shift				

Deepak Sharma
Signature of JE/SSE



लोकोमोटिव शेड साबरमती
LOCOMOTIVE Shed SABARMATI
आईओएच शेड्यूल अन्डर फ्रेम [3-फेज लोको]
SCHEDULE IOH Schedule Under Frame
[3-Phase Electric Loco]

लोको संख्या Loco No
शेड्यूल शुरू करने का दिनांक

32461

क्लास

WAG9HC

शेड्यूल

10-02-2026

शेड्यूल पूरा होने के दिनांक Date Sch completed

IOH

28-03-26

	सामान्य निरीक्षण General Inspection	अंतिम निरीक्षण Final Inspection
दिनांक Date/ पाली Shift	Date-10-02-26 Shift-8/17 h	Date-28-03-26 Shift-8/17 h
पदनाम Designation/	Mech-Ist	MCF
टिकट सं. Ticket No.	188	166 & 157
टेकनीशियन के नाम	ISHWER Parmar	ADITYA Raj & Sudhakar

ग्राहकों की प्रतिक्रिया / बुकिंग / त्रुटि संग्रह (लोको पायलट द्वारा बुक किया गया प्रतिक्रिया का रिकॉर्ड करें) Customer Feed Back/Bookings/fault archive (Record feedback booked by Loco Pilot)			
क्र सं SN	ग्राहकों की प्रतिक्रिया बुकिंग Customer Feed Back/ Bookings	कार्यवाही की Action Taken	टी सी एन का नाम द्वारा ध्यान दिया Name of TCN/Attended By

सामान्य निरीक्षण के दौरान शेड्यूल कार्य देखा Non schedule work noticed during incoming inspection			
क्र सं SN	गैर शेड्यूल कार्य का ब्यौरा Details of Non schedule work	कार्यवाही की Action Taken	टी सी एन का नाम/ द्वारा ध्यान दिया । Name of TCN/Attended By

पुनरावर्ती गैर शेड्यूल कार्य (एम आई सेल द्वारा अवगत करवाया / SLAM से नोट किया गया) Repeated Non schedule work (As advised by MI Cell/Noted from SLAM)			
क्र सं SN	गैर शेड्यूल कार्य का ब्यौरा Details of Non schedule work	कार्यवाही की Action Taken	टी सी एन का नाम/ द्वारा ध्यान दिया । Name of TCN/Attended By

आवश्यक परिवर्तनीय वस्तुओं की सूची List of Must Change items

क्र.सं. SN	पीएल सं. PL No.	वस्तु का विवरण Description of item	मात्रा Qty	TOH	IOH	स्थिति Status	टिप्पणी Remarks
1	Non stock	Knuckle, clevis, clevis pin and yoke pin		X	✓	Changd	New Fitted
2	Non stock	Striker-block wear plate		X	✓	Changd	New Fitted
3	29040127	Happy pads [3 mm]		X	✓	Changd	New Fitted
4	Non stock	Happy pads [6 mm]		X	✓	Changd	New Fitted
5	Non stock	Safety sling, pins and R- clips		X	✓	Changd	New Fitted
6	29100010	Elastic Ring and tab washers		X	✓	Changd	New Fitted
7	Non stock	Washer		X	✓	Changd	New Fitted
8	Non stock	All fasteners of Elastic ring retaining plate		✓	✓	Changd	New Fitted
9	29310088	All Steel lock Nuts for lifting of locos A Secondary Dampers (Yaw, Vertical, horizontal) B Primary Dampers top side C Traction link D Gear Case Support arm NOTE: Nylock nuts should be replaced in every overhauling		X	✓	Changd Changd ReFitted ReFitted	New Fitted New Fitted Re Fitted Re Fitted
10		All pins and pads of brake rigging		✓	✓	Changd	New Fitted
11		All bushes of brake rigging		X	✓	Changd	New Fitted
12		All rubber components and fasteners of bogie		X	✓	Changd	New Fitted
13		Brake hanger of WAP7/WAG9		X	✓	OH done	OH done Fitted
14		Safety sling of brake hanger of WAP7/WAG9		✓	✓	Changd	New Fitted
15	29250079	Axle box "O" ring and cover bolts		✓	✓	Changd	New Fitted
16		'O' ring of MSU				Changd	New Fitted
17	80012012	Gear case oil		✓	✓	Pre oil fill	Fresh oil Fill up RR 460
18	Non stock	Gearbox oil seal / "O" rings		X	✓	Changd	New Fitted
19	Non stock	Drain plug washer and oil filling plug washer		✓	✓	Attended	New Fitted
20	Non stock	Gauge glass oil seal		✓	✓	Changd	New Fitted
21	29100203	Spheriblocs of the axle guide rod (WAP7/WAG9/9H)		X	✓	Changd	New Fitted
22	29100203	Spheriblocs of the TM support arm (WAP7/WAG9/9H)		✓	✓	Changd	New Fitted
23		Spheriblocs of all types of dampers		X	✓	Changd	New Fitted

SN	Description	Incoming	Outgoing
1.	Wheel Dia/TT Date		Dt: 22-03-26
2.	Front Frame No./Axle Box Type	SL No. 341 / NDC	SL No. 341 / NDC
3.	Rear Frame No./Axle Box Type	SL No. 346 / FAG, NBC, NBC	SL No. 346 / FAG, NBC, NBC

A Bogie Disconnection and Dismantling				
SN	Details of activity	Standard value	Observed value	Name of TCN
1	Remove Traction link from bogie end and safety wire rope, Disconnect Chain link connection, secondary vertical dampers, Lateral dampers, Horizontal, YAW dampers, Pull rod pins, Earth connections, Hand brake chain, Bogie hose and sand pipe etc.	All items disconnected	All Items Disconnect	J. Shwari
2	Ensure that the SPM cable, TM cable, Bellows Body to bogie shunts etc. are disconnected before lifting	Ensured	Checked ok	Amrishi
3	Arrange for body lifting from the bogie (Care shall be taken to see that the Traction motor cables, or any other items connected between body to bogie are not getting dragged / Damaged while lifting)	Lifted	Checked ok	J. Shwari
4	Ensure while placing the loco body on stand, stands to be placed only at the nominated location under the body. Never place stands below loco cabs.	Ensured	Checked ok	Amrishi

5	All the removed equipment's and components are to be marked and serial nos. to be noted for identification.	Marked	Recorded	Bhargava
6	Shims and compensating plates with each spring to be tied along with the respective springs.	Tied	Checked	Ishwar
7	Remove torque arm, gear case etc. (IOH sch item)	Removed	Removed	Aditya
8	Remove all 6 Traction motors and handover to TM section for overhaul. (Take care to avoid spilling of oil) Note: Removal of TM is required in IOH and conditional basis in TOH for axle changing etc.	Removed	Removed	Ishwar
9	Gear case (6 Nos.) to be kept in Gear case overhauling area as set. Number the gear case. (Take care to avoid damages to the machined surfaces of gear case. Any scratches may cause for oil leakage) (IOH sch item)	Marked	Recorded	Bhargava
10	Remove all the axle guides at bogie side. (IOH sch item)	Removed	Removed	Bhargava
11	Lift the bogie frame from the wheels & keep in bogie overhauling area. (IOH sch item)	Lifted	Lifted Frame out	Ishwar
12	Segregate the springs for testing/ grouping (IOH sch item)	Segregated	done	Amrishi
13	Keep the fasteners in the nominated location.	Kept	Kept	Bhargava
14	Any abnormalities noticed in Traction motors, Bogie, Push pull rod assembly etc. to be noted down in register.	Done	done	Ishwar
15	Remove axle guides from Axle box for changing spheriblocs in IOH and on conditional basis in TOH	Removed	Removed	Bhargava
16	Chassis clean & DPT by lab.			

B	Wheel and Axle Measurement											
SN	Details of incoming activity										Action taken	Name of TCN
1.	Measure and record Wheel Diameter, Root/Flange Wear & Tread Wear										Recorded	Aditya
	Wheel No.	Wheel Tread dia	Root Wear	Flange Wear	Tread Wear	Wheel No.	Wheel Tread dia	Root Wear	Flange Wear	Tread Wear		
	1	1091.20	1	.5	.5	2	1090.78	1	.5	.5	Recorded	Aditya
	3	1090.33	1	.5	.5	4	1090.47	1	.5	.5		
	5	1091.28	1	.5	.5	6	1091.23	1.5	.5	.5		
	7	1091.17	1.5	.5	.5	8	1090.89	1	.5	.5		
	9	1091.74	1.5	.5	.5	10	1091.79	1.5	.5	.5		
	11	1091.24	1	.5	.5	12	1091.45	1	.5	.5		

2.	Measure and record Bull Gear K- Value(dimension)						Recorded	Ishwar
	Class of loco	Gear Ratio	No. of teeth	Standard	Service Limit			
	WAP-7	20:72	9	260.99 mm	260.45 mm			
	WAG-9HC	21:107	15	317.09 mm	317.17 mm	316.49		
	Axle No.	1	2	3	4	5	6	
	Actual	316.90	316.85	316.90	316.70	317.10	316.80	Recorded Sudhakar
3.	Examine all wheels visually for any crack.						Checked ok	Ishwar
4.	Examine all wheels visually for any flats.						Checked ok	Amrishi
5.	Examine all wheels visually for any cavities.						Checked ok	Sudhakar
6.	Examine all wheels visually for any metal build up.						Checked ok	Ishwar
7.	Examine all wheels visually for any scoring and grooving.						Checked ok	Sudhakar
8.	Wheel Gauge [Date measured on _____]							
	Axle No. ↓	A	B	C	D	Average	Recorded	Ishwar Sudhakar
	1	1596.32	1596.45	1596.55		1596.45		
	2	1596.35	1596.00	1596.00		1596.11		
	3	1596.45	1596.35	1596.05		1596.25		
	4	1595.95	1596.00	1596.25		1596.06		
	5	1596.40	1596.40	1596.45		1596.41		
	6	1596.55	1596.55	1596.70		1596.46		

AB-Axle Box

SN	Details of incoming activity							Action taken	Name of TCN	
9	Measure and record MSU Lateral play (in thou)							Recorded	Ishwar	
	MSU No.	1	2	3	4	5	6			
	Actual	0.027"	0.013"	0.020"	0.016"	0.015"	0.010"	Checked ok	Sudhakar	
10	Record Axle Identification No. (UST)							Recorded	Sudhakar	
	Location	1	2	3	4	5	6			
	Axle no.	2319	5401	4324	7438	4409	3037			
	Make	NBC	NBC	NBC	FAG	NBC	NBC			
11	Record Wheel Disc Identification No.							Recorded	Ishwar Sudhakar	
	Wheel No.	Wheel Disc Identification No.		Wheel No.	Wheel Disc Identification No.					
	1	12093 T2 LOS446 W544		2	12093 EOC446 W544					
	3	T2093 T2 LOB2039 W544		4	T2 EOC446 W544					
	5	12093 T2 FOM7032 W548		6	12093 T2 FOM7027 W548					
	7	12093 T2 FOG5075 W548		8	12093 T2 FM76032 W548					
	9	12093 T2 FM70020 W548		10	12093 T2 FM70032 W548					
	11	12093 T2 FMA5016 W548		12	12093 T2 FMA5006 W548					
C	Other Items									
1.	Other Items		Front		Rear		Both Side Knuckle & Pin Clavis & Pin Screw Coupling New Fitted Checked ok	Ishwar Sudhakar		
	a) Coupler Draft Condition		Checked ok		Checked ok					
	b) Cattle Guard Condition		-		-					
	c) CBC Condition		Checked ok		Checked ok					
	d) Coupling Condition		New Fitted		New Fitted					
	e) Others Remarks		-		-					
	f) Hand Brake condition		Hand brake changed		changed					

SN	Details of Works/Inspection during schedule						Action taken	Name of TCN	
D	MSU Overhauling								
SN	Details of Activity	1	2	3	4	5	6	DPT by lab - JS.Sinha Cleaned DPT by lab - JS.Sinha Checked ok DPT by lab - T. S. Sinha Bolt Torque 80 N.M Bolt Tight 80 N.M DPT by lab - T. S. Sinha New Fitted Fresh Grease Fill up.	Ishwar JS.Sinha Ishwar T. S. Sinha Activity Sudhakar T. S. Sinha Sudhakar Activity
1.	Carry out DPT of wheel disc sprang hole area	OK	OK	OK	OK	OK	OK		
2.	Carry out DPT of bull gear during TOH/IOH and in case of TM/Axle replacement. Clean gear before DPT	OK	OK	OK	OK	OK	OK		
3.	Check the condition of the TM mounting face of MSU	OK	OK	OK	OK	OK	OK		
4.	Check the MSU by DPT at vulnerable locations.	OK	OK	OK	OK	OK	OK		
5.	Tighten of allen key bolts on GE side and Hex head bolts on NGE side at 80 nm torque	OK	OK	OK	OK	OK	OK		
6.	Tighten the bolts securing the Labyrinth ring with main gear hub at 80 nm torque	OK	OK	OK	OK	OK	OK		
7.	Check the gear profile for any crack or damage by DPT during IOH sch	OK	OK	OK	OK	OK	OK		
8.	Both MSU 'O' rings new to be renewed. Ensure the joint of 'O' rings at top side.	OK	OK	OK	OK	OK	OK		
9.	Check grease sample for metal content of Axle tube at NDE side. (Shed limit: 1000 mg/100 g of grease)	OK	OK	OK	OK	OK	OK		

10	Carry out greasing in axle tube from NDE side (Quantity: 100 g)	OK	OK	OK	OK	OK	OK	Greasing done	Aditya
11	Any other remarks	OK	OK	OK	OK	OK	OK	Checked OK	Sudhakar

E Axle Box Overhauling										
SN	Details of Activity		1	2	3	4	5	6		
1.	Send sample to Lab for metal content checking (Limit: Max 250 mg/100 g grease)	GE	OK	OK	OK	OK	OK	OK	Gra Sample by. Lab - Torun	
2.	Visually inspect the axle boxes for wear/damage. Pay particular attention to the guide rod mounts and limit stops. Replace the axle box housing if defective	NGE	OK	OK	OK	OK	OK	OK	Checked OK	Aditya
		NGE	OK	OK	OK	OK	OK	OK	Checked OK	Manjara
3.	Axle Box (AB) housing (old/new)	GE	old	old	old	old	old	old	Checked OK	Aditya
		NGE	old	old	old	old	old	old		
4.	Axle box inner diameter (ID) Max.: 250.046 mm Min.: 250.000 mm	GE	250.04	250.02	250.04	250.04	250.02	250.03	Recorded	Sudhakar
		NGE	250.04	250.03	250.03	250.03	250.03	250.04		
5.	Axle Box (AB) bearing (old/new)	GE	old	old	old	old	old	old	Checked OK	Aditya
		NGE	old	old	old	old	old	old		
6.	Axle box bearing OD Max.: 250.000 mm Min.: 249.970 mm	GE	249.97	249.98	249.98	249.97	249.97	249.98	Recorded	Sudhakar
		NGE	249.98	249.97	249.97	249.97	249.98	249.97		
7.	Check the condition of inner race for any slackness / cracks/pitting	GE	OK	OK	OK	OK	OK	OK	Checked OK	ISHWARI
		NGE	OK	OK	OK	OK	OK	OK		
8.	Check the condition of abutment ring for any slackness/rubbing.	GE	OK	OK	OK	OK	OK	OK	Checked OK	Sudhakar
		NGE	OK	OK	OK	OK	OK	OK		
9.	Check condition of Rollers for pitting marks.	GE	OK	OK	OK	OK	OK	OK	Checked OK	Aditya
		NGE	OK	OK	OK	OK	OK	OK		
10.	Check condition of cage for damages	GE	OK	OK	OK	OK	OK	OK	Checked OK	ISHWARI
		NGE	OK	OK	OK	OK	OK	OK		
11.	Check condition of Free rotation of rollers.	GE	OK	OK	OK	OK	OK	OK	Checked OK	Aditya
		NGE	OK	OK	OK	OK	OK	OK		
12.	Check condition of end fittings including dust caps	GE	OK	OK	OK	OK	OK	OK	Checked OK	Sudhakar
		NGE	OK	OK	OK	OK	OK	OK		
13.	Check condition of Axle box rear cover for rubbing marks and damages.	GE	OK	OK	OK	OK	OK	OK	Checked OK	ISHWARI
		NGE	OK	OK	OK	OK	OK	OK		
14.	Check rear cover fixing bolts (M16 x 45 mm) and replace if required. Ensure tightness to 100 Nm torque	-	OK	OK	OK	OK	OK	OK	100 Nm Torque done	Aditya
15.	Replace O-rings of end cover. Apply Locktight-518 sealant.	-	New	New	New	New	New	New	New Fitted	Sudhakar
16.	Renew end plate socket head cap screw on conditional basis.	GE	New	New	New	New	New	New	New fitted	Aditya
		NGE	New	New	New	New	New	New		
17.	Tighten end plate socket head cap screw (M20x50) to 340 Nm torque.	GE	OK	OK	OK	OK	OK	OK	Tight 340 Nm end plate Bolt	Sudhakar
		NGE	OK	OK	OK	OK	OK	OK		Aditya
18.	Check front cover fixing bolts and tighten [Old/New] M16x2P45 mm x 38 thd, 8.8 Grade.Torque-100 Nm	GE	OK	OK	OK	OK	OK	OK	Front cover Bolt Torque 100 Nm	Sudhakar
		NGE	OK	OK	OK	OK	OK	OK		Aditya
19.	Renew spring washers	GE	New	New	New	New	New	New	New provided	Sudhakar

		NGE	OK	OK	OK	OK	OK	OK	Recabel	Activity
20.	Bearing make (NEI-RB 5080/ SKF-BC-20067A/FAG)	GE	NBC	NBC	NBC	FAG	NBC	NBC	Apply shell Grease	Suelhaker
		NGE	NBC	NBC	NBC	FAG	NBC	NBC		
21.	Apply grease and provide axle box with bearing on inner racers. (shell gadus) [SKF: 470 g, NEI: 650 g, FAG: 390-430 g]	GE	OK	OK	OK	OK	OK	OK		
		NGE	OK	OK	OK	OK	OK	OK		
22.	Check and record bearing clearance after mounting [Std: 0.07 mm (min)]	GE	-	-	-	-	-	-		
		NGE	-	-	-	-	-	-		
23.	Check and record radial clearance (clamping plate to end cover at 2 locations each) [Std: 0.7 mm (min)]	GE	-	-	-	-	-	-		
		NGE	-	-	-	-	-	-		
24.	Check and record gap between clamping plate inner face to axle end with special gauge [Std: 3 mm approx. (min 1.5 mm)]	GE	-	-	-	-	-	-		
		NGE	-	-	-	-	-	-		
25.	Check and record Axle box cover clearance [Std: 0.35 mm (min)]	GE	-	-	-	-	-	-		
		NGE	-	-	-	-	-	-		
26.	Axial clearance/play of bearing [SKF & NEI: 0.4 to 0.7 mm FAG: 0.2 to 0.4 mm]	GE	-	-	-	-	-	-		
		NGE	-	-	-	-	-	-		
27.	Apply Red RTV application on all covers and Earth Return Cable on AB No. 1, 6, 7, 12	-	OK	OK	OK	OK	OK	OK	Apply Red RTV All Cover Earth Return	Activity
28.	Measure the wear of carbon brush and see it does not go below the groove line. Replace if required.	-	OK	OK	OK	OK	OK	OK	Check OK	Activity
29.	Check intactness of earth return path.	-	OK	OK	OK	OK	OK	OK	Check OK	Suelhaker

F Axle box overhauling measurement record

SN	Details of Activity	Old / New	1	2	3	4	5	6		
1.	Lipped inner race	GE	New	New	New	New	New	New	New	Activity
		NGE	New	New	New	New	New	New	New	
2.	Plain inner race	GE	New	New	New	New	New	New	New	Measurement
		NGE	New	New	New	New	New	New	New	
3.	Bearing	GE	old	old	old	old	old	old	old	Suelhaker
		NGE	old	old	old	old	old	old	old	
4.	AB housing	GE	old	old	old	old	old	old	old	Activity
		NGE	old	old	old	old	old	old	old	
5.	Axle Journal dia at lipped inner race Max: 150.068 mm Min: 150.043 mm	GE	-	150.62	150.62	150.62	150.63	-	-	Recabel
		NGE	-	150.63	150.63	150.62	150.63	-	-	Suelhaker
6.	Axle Journal dia at lipped inner race (step size) Max: 148.068 mm Min: 148.043 mm	GE	-	-	-	-	148.63	148.62		Activity
		NGE	-	-	-	-	148.62	148.63		
7.	Axle Journal dia at plain inner race Max: 150.068 mm Min: 150.043 mm	GE	-	150.63	150.64	150.64	150.63	150.62	150.64	Recabel
		NGE	-	150.64	150.63	150.63	150.64	150.63	150.64	Suelhaker

8.	Axle Journal dia at plain inner race (step size) Max: 148.068 mm Min: 148.043 mm	GE	-	-	-	-	-	-	-	-	-	-
		NGE	-	-	-	-	-	-	-	-	-	-
9.	Axle Journal dia at abutment ring Max: 170.146 mm Min: 170.186 mm	GE	170.15	170.15	170.16	170.15	170.15	170.16	Recorded	Aditya Suchhaker		
		NGE	170.15	170.15	170.15	170.16	170.15	170.16				
10	Waviness (To be checked with straight edge) Std: Length of continuous line-more than 2/3 of journal length.	GE	-	-	-	-	-	-	-	-	-	-
		NGE	-	-	-	-	-	-	-	-	-	-
11	Fillet radius of thrower (More than R 8.5)	GE	-	-	-	-	-	-	-	-	-	-
		NGE	-	-	-	-	-	-	-	-	-	-
12	Lipped race ID (SKF/NEI/FAG) Max: 150.000 mm Min: 149.975 mm	GE	-	-	-	-	-	-	-	-	-	-
		NGE	-	-	-	-	-	-	-	-	-	-
13	Plain race ID (SKF/NEI/FAG) Max: 150.000 mm Min: 149.975 mm	GE	-	-	-	-	-	-	-	-	-	-
		NGE	-	-	-	-	-	-	-	-	-	-
14	Lipped race OD before fitment [175 mm, approx.]	GE	-	-	-	-	-	-	-	-	-	-
		NGE	-	-	-	-	-	-	-	-	-	-
15	Plain race OD before fitment [175 mm, approx.]	GE	-	-	-	-	-	-	-	-	-	-
		NGE	-	-	-	-	-	-	-	-	-	-
16	Lipped race OD after fitment. [As measured]	GE	-	OK	OK	OK	OK	OK	OK	Checked OK	Suchhaker	Aditya
		NGE	-	OK	OK	OK	OK	OK	OK			
17	Plain race OD after fitment. [As measured]	GE	-	OK	OK	OK	OK	OK	OK			
		NGE	-	OK	OK	OK	OK	OK	OK			
18	Interference of lipped race	GE	-	-	-	-	-	-	-	-	-	-
		NGE	-	-	-	-	-	-	-	-	-	-
19	Interference of plain race	GE	-	-	-	-	-	-	-	-	-	-
		NGE	-	-	-	-	-	-	-	-	-	-

SN	Details of Works/Inspection during schedule	Action taken	Name of TCN
6	Bogie, Dampers and Brake Rigging Maintenance Record		
1.	Clean the muck, dirt & oil etc. with scrapper & wire brush.	Cleaned	Cleaned Ishwar
2.	Examine the bogie frame and all welded components and parts for any sign of crack/damage.	Normal.	Checked OK Normal Amrisha
3.	Examine all bogies mounted equipment (including earthing straps) for damage and security of fastening.	Good	Checked OK Sunit
4.	Check the earth shunts provided on bogie	Normal	Normal Aditya
5.	Dismantle brake hangers and brake rigging components in IOH and replace damaged ones. Replace all pins, bushes and pads of brake rigging with new one in IOH and on condition basis in TOH.	Checked, Normal. New pins/ Old	Checked Normal OK and Refitted Sangeet Bhargava Ishwar
6.	Clean the brake hangers thoroughly.	Cleaned	Cleaned Ishwar
7.	Check for any bend, crack twist	Good	Checked OK Sunit
8.	Check condition of bush seating for ovality and repair them to standard size. (IOH sch item)	Good	Bushing done Amrisha

9.	Renew all bushes on brake hangers (Except Mn steel bushes) (Wear limit 1.5 mm) [IOH sch item]	New/ Old	New	Bharat
10	Check the condition of brake head for any unusual wear and replace if required. (IOH sch item)	Good	Checked OK	Aditya
11	Check condition of friction arm spring and replace if required. (IOH sch item)	New/ Old	Checked OK	Ishwar
12	Check for free movement of brake hangers in brackets.	Freely moving	Checked OK	Bharat
13	Ensure the availability of safety slings for Brake hanger. Replace safety slings on conditional basis [SMI 244]	Ensured	New	Aditya
14	Check and lubricate top slack adjusters	Lubricated	Lubricated	Ishwar
15	Apply ANR on all the thread bolts before tightening.	Applied	Applied	Aditya
16	Torque arm fixing brackets of Bogie-1&2 [TM support brackets of BG-1&2] DPT to be done.	Normal	Normal DPT by lab	Tarun
17	Pin for traction bar welding portion DPT to be carried out	Normal	DPT by lab	JS Sinha
18	DPT to be done on all damper fixing brackets, Chain link brackets etc.	No crack	DPT by lab	Tarun
19	Check all brake hangers and top brackets. Replace damaged ones.	Good	Checked OK	Bharat
20	Replace all cotter pins, split pins etc.	New	New	Aditya
21	Check the provision of stoppers and locking arrangements of all pins. (MP-MOD.VL.02-07-09)	Provided	Provided	Ishwar
22	Check condition of Horizontal damper's fixing brackets.	DPT done	DPT by lab	
23	Check condition of Yaw damper's fixing brackets.	DPT done	DPT by lab	Tarun
24	Check condition of Secondary vertical damper's fixing brackets	DPT done	DPT by lab	
25	Check condition of primary vertical damper's fixing brackets	DPT done	DPT by lab	
26	Check condition of Lateral bump stop fixing brackets.	Good	Checked OK	Sangeer
27	Check condition of Vertical bump stop fixing brackets.	Good	Checked OK	Sujit
28	Check condition of Secondary spring guides.	Good	Checked OK	Aditya
29	Check all axle guide fixing posts on bogies and carry out DPT check	Good	DPT by lab	JS Sinha
30	Check different type of dampers for oil leakage and damage to the end fittings like spheriblocs etc. overhaul/replace on condition basis.	Old/ New	New	Sangeer
31	Check condition of all brake tie bars for any crack or bend. Replace if required.	Good.	Checked OK	Sujit

H Gear Case Overhauling

SN	Details of Activity	Old / New	1	2	3	4	5	6		
1.	Inspect the gear case for any oil leakage, external hits/damage. Rectify As Required	Good old OK	OK	OK	OK	OK	OK	OK	All Gear case opened Cleaned Checked OK	Bharat Ishwar Sujit
2.	Check condition of all threaded holes.	Good old OK	OK	OK	OK	OK	OK	OK	Refitted	Bharat
3.	Replace gear case oil with new one.	5 litres (min)	7.00	7.00	7.00	7.00	7.00	7.00		
4.	Overhaul Oil gauge assembly. Replace drain plug washer and oil filling plug washer.	Good	OK	OK	OK	OK	OK	OK	Checked OK	Sangeer
5.	Overhaul breather assembly. Replace rubber gasket.	Good	OK	OK	OK	OK	OK	OK	Checked OK Oil done	Bharat
6.	Check metal content of gear case oil		OK	OK	OK	OK	OK	OK		
7.	If above 12000ppm, G/Case to be removed and checked thoroughly		OK	OK	OK	OK	OK	OK	All Gear case Refitted	Ishwar

I Assembling of Motorized Bogie and testing											
SN	Details of Activity		Old / New	1	2	3	4	5	6		
1.	Ensure MSU is overhauled and in cleaned condition. Apply adhesive on MSU.		-	OK	OK	OK	OK	OK	OK	Checked	Bharat
2.	Check dowels on MSU. Assemble TM on wheel set.		-	OK	OK	OK	OK	OK	OK	Checked	Aditya
3.	Measure and record backlash between bull gear and TM pinion. Standard: 0.25 to 0.46 mm, Service limit: 1.0 mm		-	OK	OK	OK	OK	OK	OK	Recorded	Ishwar
4.	Provide zyglow tested Axle cap bolts (M30x140 mm).		-	OK	OK	OK	OK	OK	OK	Zyglow Tested bolts Fitted	Aditya
5.	Condition of M30 cap bolt washer		-	OK	OK	OK	OK	OK	OK	New Fitted	Sudhakar
6.	Tighten the axle cap bolts (M30x140 mm) and torque to 1250 Nm		-	OK	OK	OK	OK	OK	OK	Tight done	Aditya
7.	Apply sealant gasket on machined surfaces of gear case and labyrinth ring. Ensure all the 3 "O" rings are replaced.		-	OK	OK	OK	OK	OK	OK	Changed	Sudhakar
8.	Assemble gear case.		-	OK	OK	OK	OK	OK	OK	Assembly	Bharat
9.	Check gear case fasteners for tightness at prescribed torque value.		-	OK	OK	OK	OK	OK	OK	Checked OK	Ishwar
	SN	Bolt size	Torque value								
	1	M16 x 200	205 Nm							All Bolt	Ishwar
	2	M12 x 160	80 Nm							Torque	Singh
	3	M12 x 80	80 Nm							done	Bharat
	4	M12 x 70	80 Nm								
	5	M12 x 30	80 Nm								
10	Ensure provision of rubber/copper gasket on the inner allen screw.		-	OK	OK	OK	OK	OK	OK	Checked OK	Ishwar
11	Conduct running test of TM and gear case assembly for one hour in each direction.		-	OK	OK	OK	OK	OK	OK	one hour Motoring done	Singh
12	Measure and record rise of temperature (above ambient) at following locations		-	OK	OK	OK	OK	OK	OK	Recorded	Singh
a)	Axle box - GE (SMI/246)		-	31.70	33.40	33.80	29.10	30.10	30.80		
b)	Axle box- NGE (SMI/246)		-	31.80	30.50	33.70	29.20	29.60	30.30		
c)	Gear case top		-	31.80	33.20	33.40	29.20	29.60	30.30	Checked	Aditya
d)	Gear case bottom		-	31.80	33.20	33.40	29.20	29.60	30.30		
e)	MSU - GE side		-	31.80	33.50	33.60	29.10	29.50	30.60		
f)	MSU - NGE side		-	31.90	33.40	33.50	29.10	29.60	30.30	Recorded	Singh
g)	TM-PE side		-	31.70	33.30	33.20	29.30	29.50	30.70		
h)	TM-CE side		-	31.70	33.30	33.20	29.30	29.50	30.60		
	Temperature range/limit Ambient Temp (ta) =		-	-	-	-	-	-	-		
	SN	Sr. No.	Limit								
	1	a) to d)	25 °C + ta								
	2	e) & f)	30 °C + ta								

	Details of Activity	Old / New	Bogie-1	Bogie-2		
13	Check condition of Spheriblocs and assemble guide rods on axle boxes.	-	New	New	New Fitted	Sangeer
14	Tighten and Torque the neck down and M24x300 mm bolts Torque value Neck down- 670 Nm M24x300 – 665 Nm	-	Torque done	Torque done	Torque done	Bhazad
15	Renew steel lock nuts.	-	New	New	New Fitted	Aditya
16	Clean & examine primary springs for any sign of crack/ damages	-	Ser. Tested	Ser. Tested	Ser. Tested	Aditya
17	Tested and grouped springs to be provided in IOH schedule	-	grouped	grouped	grouped	Aditya
18	Provide Primary springs with corresponding shims if any	-	provided	provided	provided	Manraj
19	Assemble Bogie on Axle	-	Assemble	Assemble	Assemble	Aditya
20	Provide torque arm. Ensure the condition of spheriblocs. Apply thread sealant. Tighten and torque the fixing bolts with nut to 665 Nm torque.	-	New	New	New Fitted	Aditya Manraj
21	Provide all primary dampers. Record condition of dampers i.e. Old/New	-	New	New	New Fitted	Sangeer
22	Provide links on middle axles	-	New	New	New Fitted	Aditya
23	Provide brake shoes	-	provided	provided	provided	Bhazad
24	Provide all bottom pull rods and modified pins and J clamps.	-	New	New	New Fitted	Ishwar
25	Check for proper seating of brake shoes on wheel tread portion	-	provided	provided	provided	Sangeer
SN	Details of Works/Inspection during schedule				Action taken	Name of TCN
J	Torque Arm, Traction Link Assembly					
SN	Details of Activity	Old / New	Cab-1	Cab-2		
1.	Check the push/pull rod (Traction link), pivot head for damage any sign of crack with DPT (at both bogie and body side)	-	old	old	old fitted	Aditya
2.	Check the intactness/security of the safety sling [Must change item during IOH].	-	New	New	New Fitted	Manraj
3.	Check the intactness of fasteners between traction rod and pivot head at body end. Apply thread sealant and tighten at 273 Nm torque. (TC/142)	-	Torque	Torque	Torque done	Aditya
4.	Replace all Steel-lock Nuts and disc lock washers of traction links at body side (if removed).	-	New	New	New fitted	Manraj
5.	Check intactness/Security of inner and outer retaining ring, bolt and plate and ensure its tightness at required torque.	-	Torque	Torque	Torque done	Sadhakar
	SN Bolt of Torque value					
	1 Outer ring 80 Nm					
	2 Inner ring 80 Nm					

6.	Check Elastic ring on body and bogie for any crack or shifting. Replace if defective and during 2 nd MOH [SMI/259]	-	New	New	New Fitted	Bharadwaj
7.	Carry out DPT check on Pivot pin welding portion.	-	OK	OK	DPT by lab -	Charanish
8.	Examine limit chain and links wear and damage.	-	New	New	New Fitted	Aditya
9.	Check condition of Spheriblocs of all torque arms. Replace if required [Must change item during IOH].	-	New	New	New Fitted	Bharadwaj
10	Check condition of Spheriblocs of all axle guide rods. Replace if required [Must change item during IOH].	-	New	New	Fitted	Singh

SN	Details of Works/Inspection during schedule	Action taken	Name of TCN
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K Centre Buffer coupling (CBC) Maintenance

1.	CBC Assembly and Side Buffers		
	a) Clean all components viz. coupler body, Knuckle, Clevis, Knuckle and Clevis and Transition screw coupling.	Removal	ISHWAR
	b) Carry out DPT on Coupler body, Knuckle, Clevis, Knuckle and Clevis pin and MPT on Transition screw coupling	New Fitted	Amish
	c) Carry out DPT on the Side Buffer foundation bolt holes for cracks	DPT lab -	JS Sinha

2.	Screw coupling work		
	Check transition screw coupling for its conditions. Check its thread and shackle for wear and record below. Punch SBI No. (ExSBI/PIC Year/Sr. No.)	New Fitted	JS Sinha

3.	Dimension and Measurement Records of CBC [Type of CBC: E type]				
SN	Parameter	Standard Value	MME to be used	Observed Value	
				Cab-1	Cab-2
1	Striker face casting to coupler horn gap	108 ± 25 mm	Outer calliper & scale	Checked OK	Checked OK
2	Distance between nose of knuckle and guard arm	130 mm (max)	Gauge No. 2	Checked OK	Checked OK
3	Knuckle pin diameter Condition: Old/New	38.0 to 41.28 mm	Vernier calliper	41.20	41.20
4	Clevis pin diameter Condition: Old/New	36 to 38 mm	Vernier calliper	38.00	38.00
5	Check slackness of Draft gear.	No slackness	Visual	Checked OK	Checked OK
6	Knuckle nose wear gauge must not pass	Not passing	Gauge 1	Checked OK	Checked OK
7	Knuckle stretch limit - Gauge must pass	Passing	Gauge 1	Checked OK	Checked OK
8	Shank wear plate thickness	New: 6 mm Wear: 5 mm (max)	Vernier	5.00	5.00
9	Striker casting wear plate thickness	New: 8 mm Wear: 6 mm (max)	Vernier	7.00	7.75
10	Articulated lock lift assembly	No crack/damage	Visual	Checked OK	Checked OK
11	Main anti creep protection	Working.	Crowbar	Checked OK	Checked OK
12	CBC operating handle	Working	Visual	Checked OK	Checked OK
13	Check condition of cattle guard for crack/deformations.	Good	Visual	Checked OK	Checked OK
14	Check condition of yoke, CBC body, draft gear and other components	Good/No crack	Visual/DPT	DPT by lab -	JS Sinha
15	Check Screw coupling condition by MPT and thread condition.	D shackle ear: 3 mm (max)	Visual	New Fitted	New Fitted
16	Raise screw coupling, it should freely raise in knuckle and clevis in closed condition.	Freely moving	Operation	Checked OK	Checked OK
17	Draft gear condition	Good	Visual	Checked OK	Checked OK
18	Condition of welding of stopper plates in body pocket and measurement	Ensure 625 mm	Visual	Checked OK	Checked OK
19	Check for lifting of lock in CBC locked position	Max- 20 mm pprox.)	Visual	Checked OK	Checked OK
20	Check free lateral movement of shank	Normal	By hand	Checked OK	Checked OK

21	Check condition of rivets on knuckle pin	Good	Visual	Not fill	Not fill
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SN	Details of Works/Inspection during schedule			Action taken	Name of TCN
Centre Buffer coupling (CBC) Maintenance					
4.	Dimension and Measurement Records of CBC [Type of CBC: H type]				
SN	Parameter	Standard Value	MME to be used	Observed Value	
				Cab-1	Cab-2
1	Knuckle pin diameter Condition: Old/New	Must pass	Go Gauge		
		Must not pass	No Go Gauge		
2	Clevis pin diameter Condition: Old/New	44.15 to 47.15 mm	Vernier caliper		
3	Check slackness of Draft gear.	No slackness	Visual		
4	Shank wear plate thickness	New: 6 mm Wear: 5 mm (max)	Vernier		
5	Double rotary lock lift assembly	No crack /damage	Visual		
6	CBC operating handle	Working	Visual		
7	Check condition of cattle guard for crack/ deformations.	Good	Visual		
8	Check condition of yoke, CBC body, draft gear and other components	Good/No crack	Visual/DPT		
9	Check Screw coupling condition by MPT and thread condition.	D shackle wear: 3 mm (max)	Visual		
10	Raise screw coupling, it should freely raise in knuckle and clevis in closed condition.	Freely moving	Operation		
11	Draft gear condition	Good	Visual		
12	Condition of welding of stopper plates in body pocket and measurement	Ensure 625 mm	Visual	NA	NA
13	Check for lifting of lock in CBC locked position	Max- 20 mm (approx.)	Visual		
14	Check free lateral movement of shank	Normal	By hand		
15	Check condition of rivets on knuckle pin	Good	Visual		
16	Check contour with knuckle profile gauge	Must pass	Inspector contour gauge-I (31000)		
17	Check the condition of the coupler by condemning limit gauge between nose of knuckle & Guard arm	Condemning Limit Gauge	Condemning Limit Gauge		
18	In case condemning limit gauge passes: - Replace knuckle and again check - If it passes again, replace the lock, and again check with gauge - If it passes again, replace the coupler body	Must not pass	Condemning Limit Gauge		
19	Check contour with contour condemning limit gauge	Must not pass	Gauge No. 34100-1		
20	Check condition of coupler device with aligning wing limit gauge	Seat & pass	Gauge No. 32600		
21	Check Knuckle nose wear with knuckle nose wear & and stretch limit gauge	Point D must not touch when A, B & C seated condition. Clearance should be more than 6.35 mm	Gauge No. 341002A		
22	Check the gap between the knuckle and lock in closed condition	Max 3.175 mm	Scale		

Centre Buffer coupling (CBC) Maintenance

Dimension and Measurement Records of CBC [Type of CBC: H type]

SN	Parameter	Standard Value	MME to be used	Observed Value	
				Cab-1	Cab-2
23	Check anti-creep protection	Working	Pry bar and screw driver		
24	Check vertical height aligning wing pocket and guard arm with Go & No-Go gauge)	Must pass	Go gauge (34101-4)		
		Must not pass	No-Go Gauge (34250-5)		
25	Check condition of compression springs and supporting device.	Good	Visual	N/A	N/A
26	Ensure availability of check nut in supporting device	Available	Visual		

5. Side Buffer

SN	Parameter	Standard Value	Observed Value			
			Buffer-1	Buffer-2	Buffer-3	Buffer-4
1	Check side buffer body for any crack.	No crack	OK	OK	OK	OK
2	Check buffer for slackness. Replace if found slack or damaged.	No slackness	OK	OK	OK	OK
3	Check length of the buffers. If less, change the buffer complete.	615 to 635 mm	642	641	642	640
4	Check fixing bolts, Castle nuts, and split pins and replace defective ones	Good	Attached	Attached	Attached	Attached
5	Replace defective side buffers.	New/Old	Checked OK	Checked OK	Checked OK	Checked OK
6	Any other remarks		-	-	-	-

SN	Details of Works/Inspection during schedule	Action taken	Name of TCN
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L Suspension (Helical Springs)

1. Clean Helical coil springs of locos and examine for cracks/ breakage.

Check & record free height as well as compressed height under load. Colour code of Helical Coil Springs fitted:

Bend I

Bogie-1 (Cab-I)

No. All Coil Spring Ser. Tested Provided by Aditya

Primary helical spring (inner) Static load: 0.948 T

Primary helical spring (Outer) Static load: 3.192 T

Location	Outer			Inner		
	Make/Year	Free height 236.40- 243.80 mm	Loaded height 190.00-194.00 mm	Make/Year	Free height 237.80- 245.20 mm	Loaded height 186.00-190.00 mm
	Spring Tested OK			Spring Tested OK		
1	02-03-26	243	191	02-03-26	241	187
2	25-02-26	239	190	02-03-26	241	187
3	02-03-26	242	191	25-02-26	241	187
4	25-02-26	241	191	25-02-25	241	187
5	02-03-26	242	190	25-02-26	241	187
6	25-02-26	241	191	25-02-26	240	187
7	02-03-26	242	191	27-08-25	241	187
8	25-02-26	241	191	25-02-26	240	187
9	02-03-26	242	191	22-12-25	241	187
10	25-02-26	241	191	05-02-25	238	187
11	02-03-26	241	191	22-12-25	240	187
12	26-02-26	240	190	02-03-26	240	187

Secondary (helical Spring)

Secondary helical spring (inner) Static load: 14.7 KN

Secondary helical spring (Outer) Static load: 96.8 KN

Outer			Inner		
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Location	Make/Year	Free height 725.90-742.20 mm	Loaded height 570.00-580.00 mm	Make/Year	Free height 486.09 -498.00 mm	Loaded height 401.50-408.50 mm
1	Spring Tested Date 06-01-26	732	578	Spring Tested Date 07-01-26	493	408
2	06-01-26	738	580	07-01-26	492	406
3	06-01-26	737	578	07-01-26	492	406
4	03-03-26	735	580	07-01-26	495	408

Bogie-2 (Cab-II)

No.	Primary helical spring (inner) Static load: 0.948 T Primary helical spring (Outer) Static load: 3.192 T					
	Outer			Inner		
Location	Make/Year	Free height 236.40- 243.80 mm	Loaded height 190.00-194.00 mm	Make/Year	Free height 237.80- 245.20 mm	Loaded height 186.00-190.00 mm
13	Spring Tested Date 25-02-26	241	191	Spring Tested Date 25-02-26	239	187
14	25-02-26	242	191	02-03-26	241	187
15	25-02-26	241	191	25-02-26	241	187
16	26-02-26	242	191	02-03-26	241	187
17	25-02-26	240	191	25-02-26	241	187
18	26-02-26	241	191	05-02-26	241	187
19	25-02-26	241	191	25-02-26	240	187
20	02-03-26	242	191	21-11-25	240	187
21	25-02-26	242	191	02-03-26	241	187
22	02-03-26	242	191	27-08-25	240	187
23	25-02-26	242	191	02-03-26	241	187
24	02-03-26	242	191	27-08-25	240	187

Secondary (helical Spring)

	Secondary helical spring (inner) Static load:14.7 KN Secondary helical spring (Outer) Static load:96.8 KN					
	Outer			Inner		
Location	Make/Year	Free height 725.90-742.20 mm	Loaded height 570.00- 580.00 mm	Make/Year	Free height 486.09 -498.00 mm	Loaded height 401.50-408.50 mm
5	Spring Tested Date 06-01-26	732	580	Spring Tested Date 07-01-26	496	408
6	06-01-26	735	579	30-01-26	490	407
7	06-01-26	737	580	07-01-26	490	406
8	03-03-26	735	580	07-01-26	499	407

SN	Details of Works/Inspection during schedule				Action taken	Name of TCN
M	Hand Brake and Sand Gear					
1.	Hand Brake					
SN	Parameter	Standard Value	Observed value			
1	Clean the hand brake assembly	Cleaned	Cleaned	Cleaned	He mand	
2	Check all components and replace defective ones.	Good	Attended	Attended	Mathiasyn	
3	Lubricate pinion, chain pulley etc.	Lubricated	Lubricate	Lubricate	mathiasyn	
4	Ensure the availability of locking bolt in chain to avoid entangling on under gear equipment	Provided	Provided	Provided	Suthakes	
5	Provide support guide for chain against infringements in between pulley and bracket	Provided	Provided	Provided	mathiasyn	
6	Check operation of handbrake for proper working	Working properly	Checked OK	Checked OK	Suthakes	

7	Paste the operating instruction for hand brake operation	Pasted.	Pasted	Pasted	Gurthakor
8	Type of mounting	Horizontal/ Vertical	Vertical	Vertical	

N

Loco Lowering / Bogie IN and Connection

SN	Parameter	Standard Value	Observed value	Ready	
1	Ensure loco body is ready for lowering.	Ready	Ready Bogie In	Bogie provided	Grinder & Manaraj
2	Ensure Vertical dampers, Yaw dampers and Lateral dampers are available in good condition.	Good	New Fitted	New Fitted	Singh
3	Ensure both the bogies are ready for lowering. Place both bogies at lowering pit line.	Ready	Bogie provided	Bogie provided	Singh
4	Check availability of traction link, pivot mounting assemblies on both ends in good condition.	Good	Good	Checked ok	Aditya
5	Check availability on bogie to body connecting chain with pin in good condition at 04 locations.	Good	Good	Checked ok	Sudhakar
6	Load secondary helical springs with corresponding shims and happy pads. (Before loading, ensure springs are cleaned, painted, and tested)	Loaded	Cleaned-Tested Painted Checked ok	Checked ok	Aditya
7	Check availability of lateral and vertical rubber stops.	Available	Checked ok	Checked ok	Singh
8	Lower loco body on bogies. Care to be taken to avoid damages to traction links	Lowered	Loco lowered	Loco lowered	Singh

SN Details of Works/Inspection during schedule

Action taken

Name of TCN

Loco Lowering / Bogie IN and Connection

SN	Parameter	Standard Value	Observed value		
9	Ensure proper seating of all helical springs	Ensured	Checked ok	Checked ok	Aditya
10	Ensure no homing of helical springs.	No homing.	Checked ok	Checked ok	
11	Ensure vertical and lateral clearances are within limit.	Within limit	Attended	Attended	
12	Provide all mechanical connections viz. Traction link, safety wire ropes, Chain link connection, secondary vertical dampers, Lateral dampers, Horizontal YAW dampers, Pull rod pins, Earth connections, Hand brake chain, Bogie hose and sand pipe etc.	Torque value: - Traction link-273 Nm H/V damper-192 Nm Yaw damper-300 Nm	Checked and Torque done	Checked and Torque done	Aditya Bharat
13	Ensure proper provision of all split pins and cotter pins.	Good	Checked ok	Checked ok	Aditya
14	Ensure availability of all safety wire ropes.	Available	New Fitted	New Fitted	Sudhakar
15	Any oscillation observed on high Speed		Checked ok	Checked ok	Sudhakar
16	Any unusual sound from under truck		Checked ok	Checked ok	
17	Check for any oil leakages.	No leakage.	Checked ok	Checked ok	

O Bogie sizes and Measurements

1	Measure and Record the following Bogie measurements data												Attended	Aditya
1	Primary Vertical clearance between axle box and bogie frame [Standard: 45 to 30 mm, Service Limit: 27 to 45mm]													
	1	2	3	4	5	6	7	8	9	10	11	12		
	35.20	33.80	32.60	32.30	37.30	29.30	30.00	29.20	32.40	35.30	35.70	30.50	Attended	Sudhakar
2	Primary Lateral clearance between axle box and bogie frame [Standard: 15 to 19 mm, Service Limit: 22 mm]													
	1	2	3	4	5	6	7	8	9	10	11	12		
	16.45	16.70	17.75	15.80	16.75	15.60	16.45	16.70	17.40	15.60	16.25	17.90	Rechecked	Aditya
3	Secondary Vertical clearance between bogie step and under frame body [Standard: 35 to 60 mm, Service Limit: 32 mm]													
	Cab-1						Cab-2							
	LP Side			ALP side			LP Side			ALP side				
	52.50			50.70			42.20			49.60				

		4	Secondary Lateral clearance between bogie step and under frame body [Standard: 45 to 50 mm, Service Limit: 55 mm]				Recorded	Arbitary
		Cab-1		Cab-2				
		LP Side	ALP side	LP Side	ALP side			
		46.60	45.10	46.50	47.70			

SN	Details of Works/Inspection during schedule							Action taken	Name of TCN
Bogie sizes and Measurements									
2	Measure and Record Side Buffer length & height, Centre Buffer height, Rail guide height etc.							Adjusted Recorded	Aditya Suchak
SN	Parameter	Standard (mm)	Service limit (mm)	Cab-1		Cab-2			
				LP Side	ALP side	LP Side	ALP side		
1	Side Buffer height	1105	1030	1104	1103	1105	1100		
2	Side Buffer length	635 ^{+2.0 / - 1.5}	615	642	641	642	640		
3	Rail guard height	119	102	115	115	111	112		
4	CBC height (E type)	1105 To 1080	1050	1103		1084			
3	Adjust brake strokes								
SN	Parameter	Standard (mm)	Observed (mm)				107 to 117 mm	10 mm (3/8",	
			1	2	3	4			
1	Brake Piston travel	R/S	107	107	109	111			
		L/S	110	107	109	107			

Remarks if any

P	नीचे दिए सेफ्टी आइटम चेक शीट जाँच करें और भरें Check and fill up the check sheet for safety items as given below		
क्र.सं. SN	सेफ्टी आइटम की विवरण Description of Safety Items	सीबी CB	दृष्टव्यणी Remarks
1.	ढीला, गायब एवं टूटे हुए अंग/ घटक । Loose, missing and broken components/parts.	✓	OK
2.	क्लैम्पिंग एवं इनके वैल्विंग/बांधनेवाला पदार्थ की दशा Clamping and condition of its welding/fastener	✓	OK
3.	ज़रूरत से ज्यादा गरम /असामान्य घघसाव भागों । Overheated / Abnormal wear in parts.	✓	OK
4.	दोनों ट्रांजिशनन स्कू कपलिंग की क्रियाशीलता का जाँच करें । Check operation of both Transition screw couplings	✓	OK
5.	सीबीसी की क्रियाशीलता आपरेटिंग रॉड तथा कपलर लॉक पपन की जाँच करें । Check operation of CBC, Operating Rod and coupler lock Pin.	✓	OK
6.	सीबीसी के ऊपर और नीचे का लाइनर की जाँच करें । Check the CBC top and bottom liner.	✓	OK
7.	सीबीसी नकल की क्रियाशीलता की जाँच करें । Check operation of CBC Knuckles	✓	OK
8.	सीबीसी को बनाए रखने (retaining) के पिन एवं सपोर्ट प्लेट बोट की जाँच करें । Check CBC retaining pin and support plate bolts.	✓	OK
9.	कैटल गार्ड तथा बैक सपोर्ट के बोट की जाँच करें । Check the bolts of Cattle guards and back supporters.	✓	OK
10.	रेल गाइड का फिटमेंट एवं उसकी ऊंचाई की जाँच करें । Check proper fitment of Rail Guard & its height.	✓	OK
11.	हॉर्न स्टे के प्लेट बोट की जाँच करें । Check Horn stay plate bolts.	✓	OK
12.	ब्रेक शू एवं ब्रेक हेंगरपिन / बोल्ट की जाँच करें । Check Brake Shoe and Brake hanger Pin/Bolts.	✓	OK
13.	ऐक्सल बाक्स के कवर और सीलिंग की जाँच करें । Check Axle box cover & sealing.	✓	OK
14.	सेफ्टी क्लैप की जाँच करें तथा यह सुरक्षा के लिए ब्रेक एडजस्टिंग पुल रॉड के साथ लगा है । Check Safety 'U' clamps and ensure that it retained brake adjusting pull rod.	✓	OK
15.	ब्रेक क्रॉस टाई बार के लिए सपोर्ट व्यवस्था की जाँच करें । Check the support arrangement for brake cross tie bars	✓	OK
16.	ब्रेक ब्लॉक एवं उसके चाबी का सही फिटमेंट की जाँच करें । Check proper fitment of Brake block & its Key.	✓	OK

	17.	सुनिश्चित करें कि ब्रेक रिलीज की स्थिति में ब्रेक ब्लॉक व्हील के साथ रगड़ नहीं रहे। Ensure that brake block does not rub with the wheel in brake release position.	✓	OK
	18.	सुनिश्चित करें कि ब्रेक लगाया स्थिति में ब्रेक पुल रॉड नहीं हिले। Ensure that brake pull rod don't move in brake applied position.	✓	OK
SN	क स SN	सेफ्टी आइटम की वववरण Description of Safety Items	सीबी CB	टिप्पणी Remarks
	19.	गियर केस माउंटिंग बोल्ट तथा सी क्लैप बोल्ट की जाँच करें। Check Gear Case mounting bolts and 'C' Clamp Bolts.	✓	OK
	20.	सस्पेंशन बेयरिंग कैप बोल्ट सीलिंग शलंग की जाँच करें। Check Suspension bearing cap bolts sealing.	✓	OK
	21.	टीएम सस्पेंशन टेपर रोलर बियरिंग ढाँचा में दरार की जाँच करें। Check for crack in the TM Suspension taper Roller bearings housing	✓	OK
	22.	हाथ ब्रेक की क्रियाशीलता सुनिश्चित करें। Check the operation of hand brake.	✓	OK
	23.	एक्सल बॉक्स तथा बोगी फ्रेम लाइनर की जाँच करें। सुनिश्चित करें सभी लाइनर आपनी जगह में हैं और वहाँ कोई ब्रेक नहीं है। Check Axle box and Bogie Frame liner. Ensure all liners are in place and there is no weld crack.	✓	OK

प्रमाणित किया जाता है कि उपरोक्त मदों को अधोहस्ताक्षरी द्वारा व्यक्तिगत रूप से जाँच किया गया है।

Certified that the above items have been personally checked by the undersigned

R	निम्न टेबल में किए गए सभी शैड्यूल का रिकॉर्ड करें [यदि आवश्यक हो तो अतिरिक्त शीट जोड़ें] Record all Modifications carried out in the following table [If required add extra sheet(s)] जोड़ा गया शीट की संख्या No. of sheet(s) added _____				
	क्र सं SN	दिनांक Date पाली Shift	मॉडिफिकेशन का विवरण Details of Modifications	टी सी एन का नाम Name of TCN	पर्यवेक्षक का हस्ताक्षर Sign of concern JE/SSE

S

निम्न टेबल में किए गए सभी गैर शेड्यूल कार्य का रिकॉर्ड करें [यदि आवश्यक हो तो अतिरिक्त शीट जोड़ें]

Record all Non-schedule work in the following table [If required add extra sheet(s)]

जोड़ा गया शीट की संख्या No. of sheet(s) added

क्र सं SN	दिनांक Date	किए गए कार्य का विवरण Work Done	टी सी एन का नाम Name of TCN	शेष कार्य Work Balance	पर्यवेक्षक का हस्ताक्षर Sign of concern JE/SSE
	पाली Shift				

अन्य किसी भी जानकारी Any Other Information

Signature of SSE /UF

Signature of SSE/JE/MCF

Siddhanta
Shri Sushratar Joshi
MCF Checked by Mech.: MCF



Section: Air Brake

CHECK LIST OF ELECTRIC LOCOMOTIVES UNDER TOH/IOH SCHEDULE

Loco type: - WAG-9HC

Out Date:- _____

Schedule: - IOH

Check Driver booking & fault Archive:

SN.	Repair	Action Taken	Technician

IRC Book Repairs :

[illegible]

LOCO NO:- SCHEDULE:-		STANDARD		IRC Date:		FRC Date:																														
Sr. No	Testing Parameters			Sign. Of staff : <i>प्रीति</i>		Sign. Of staff : <i>Sahish.sam.</i>																														
				Sign. Of Sup.:		Sign. Of Sup.:																														
1	Aux. comp. loading time (0 to 8 kg/cm2)	CCB – 12 sec E-70 -60 sec		1.20 sec		59 sec.																														
2	Auxiliary Manifold PAN 1&2 Magnet Valve Leakage after raising pantograph.	0.4kg/cm2 in 5 min.		PT1	PT2	PT1	PT2																													
3	MAIN COMPRESSOR a)Timing for CP to build from 0 to 10 Kg/cm2 (Each CP)	4 to 7 Min.		CP 1	CP 2	CP 1	CP 2																													
	b)Timing for CP to build from 0 to 10 Kg/cm2 (Both CP)	3.5 Min/Max.		3.24 sec	3.24 sec	5.45 sec	6.83 sec																													
4	Timing for CP to build from 8 to 09.5 Kg/cm2	30 to 40 sec Max.120 sec		CP 1	CP 2	CP 1	CP 2																													
						29 sec.	35 sec																													
VIBRATION CHECK				Vibration (In mm)																																
				CP-1		CP-2																														
<table border="1"> <thead> <tr> <th>Sr. No.</th> <th>Std. Value In mm PEAK-PEAK</th> <th></th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0.001 to 4.0 mm</td> <td>Mount CP END (V)</td> </tr> <tr> <td>2</td> <td>0.001 to 4.0 mm</td> <td>Mount 2 Motor End outside (V)</td> </tr> <tr> <td>3</td> <td>0.001 to 4.0 mm</td> <td>Mount 2 Inside (V)</td> </tr> <tr> <td>4</td> <td>0.001 to 4.0 mm</td> <td>Crank Case (V)</td> </tr> <tr> <td>5</td> <td>0.001 to 4.0 mm</td> <td>Motor Body (V)</td> </tr> <tr> <td>6</td> <td>0.001 to 4.0 mm</td> <td>Motor Body (H)</td> </tr> </tbody> </table>				Sr. No.	Std. Value In mm PEAK-PEAK		1	0.001 to 4.0 mm	Mount CP END (V)	2	0.001 to 4.0 mm	Mount 2 Motor End outside (V)	3	0.001 to 4.0 mm	Mount 2 Inside (V)	4	0.001 to 4.0 mm	Crank Case (V)	5	0.001 to 4.0 mm	Motor Body (V)	6	0.001 to 4.0 mm	Motor Body (H)	1	2	3	4	5	6	1	2	3	4	5	6
Sr. No.	Std. Value In mm PEAK-PEAK																																			
1	0.001 to 4.0 mm	Mount CP END (V)																																		
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5	0.001 to 4.0 mm	Motor Body (V)																																		
6	0.001 to 4.0 mm	Motor Body (H)																																		
				1.3	0.6	0.5	0.4	0.1	1.7	1.3	0.5	0.5	0.3	0.1	1.2																					
5	AIR DRYER																																			
	Working of air dryer Tower to change in every Min. (FTIL,SIL) In Two Min. (KBIL.) As per OEM's	Ok/ Not Ok		TRD OK		TRD OK																														
	Check Air dryer Bypass Cock A,B & C Working & Color code.	Ok/ Not Ok		OK		OK																														
	Humidity Indicator color	Blue/ White																																		
	Humidity and Dew Point Meter Reading to Check	SMI No. RDSO/2024/EL/SMI/0xxx, Rev. '0', Dated 06.03.2025 RH - Below 35 %		RH 017.237 TEMP 025.4 C D.P -01.0 mois 04847 ppm		RH 004.954 Y. TEMP 031.4 C D.P -12.6 C mois 01985 ppm																														
	Air dryer charging when compressor is working (FTIL 120sec/SIL 180 Sec.)	TC No. RDSO/2011/EL/TC/0108, Rev.'o', Date 07-03-2011		—		—																														
6	Working of Un-loader	Ok/ Not Ok		OK		OK																														
7	Working of auto drain valve and manual Draining	Ok/ Not Ok		OK		OK																														

Brake Test by A-9	BP	BC Pr.	CAB-1		CAB-2		CAB-1		CAB-2	
			BP	BC	BP	BC	BP	BC	BP	BC
Running	5.5 ± 0.07	0.0	5.0	0.00	5.0	0.0	5.0	0.0	5.0	0.0
Release	5.0 ± 0.07	0.0	4.7	0.05	4.7	0.05	4.5	0	4.7	0.4
Minimum Service	0.40 reduction	0.4 ± 0.10	3.4	2.5	3.4	2.5	4.7	0.4	3.4	2.5
Full Service	1.65 reduction	2.5 ± 0.10	1.6	2.5	1.6	2.5	8.4	2.5	1.6	2.5
Emergency	0	2.5 ± 0.10	0	2.5	0	2.5	0	2.5	0	2.5

09	Check A9 Brake Cyl. Pressure (Co-action)	2.5 ± 0.1	CAB-1	CAB-2	CAB-1	CAB-2
			2.5	2.5	2.5	2.5
10	Check SA-9 Brake Cyl. Pressure	3.5 ± 0.2 kg/cm2	CAB-1	CAB-2	CAB-1	CAB-2
				3.5		3.5
11	Check Auto Brake Timings Application up to 95% BC & Release up to 0.4 Kg/cm2	Apply time 18 to 24 Sec. Release time 45 to 60 Sec. For WAG-9	Appl.	Rel.	Appl.	Rel.
			16 sec	48 sec	15 sec	50 sec
12	Check Direct Brake (SA9) Timing	Apply time. 8 sec. Release time 8-12 Sec	Appl.	Rel.	Appl.	Rel.
			8 sec	12 sec	8 sec	12 sec
13	Check BP leakage (Isolate cock 70) E-70 EBV & LT Switch	0.7 kg/cm2/ 5min	NO DROP		NO DROP	
14	Check BP drop by emergency cock (RAS Valve) ALP side	OK/ Not Ok	CAB-1	CAB-2	CAB-1	CAB-2
			OK	OK	OK	OK
15	Check Leak hole test by 7.5 mm leak hole	BP should not fall below 4.0 kg/cm2 in 60 sec.	CAB-1	CAB-2	CAB-1	CAB-2
			4.8	4.8	4.8	4.8
16	Check a) LPO working put (DBC to release mode)	BP raise to 5.5 ± 0.2 kg/cm2	CAB-1	CAB-2	CAB-1	CAB-2
			—	—	—	—
	Check b) LPO Timing	BP stable for 60 to 80 sec fall to 5.0 kg/cm2 in 180 + 20 sec.	65 sec		60 sec	
17	Check a) FP pressure Setting	6.0 ± 0.2 kg/cm2	CAB-1	CAB-2	CAB-1	CAB-2
			6.0	6.0	6.0	6.0
	Check b) FP Leak hole drop (on 5.5 mm leak hole)	1.5 kg/cm2 in 1 min.	5.8	5.8	5.8	5.8
	Check c) FP Pressure Drop (leakage check) (Isolate cock 136)	0.7 kg/cm2 in 5 min.	NO DROP		NO DROP	
18	Check MR leakage check after SA9 in applied condition	0.5 kg/cm2 in 5 min.	No Leage		NO leakage	
19	Check Working of PVEI (Foot Pedal) (Quick Release)	OK/ Not Ok	CAB-1	CAB-2	CAB-1	CAB-2
			OK	OK	OK	OK
20	Check Working of horns.	OK/ Not Ok	CAB-1	CAB-2	CAB-1	CAB-2
			OK	OK	OK	OK

21	a) Setting of MR Safety Valves.(check sealing)	MR 10.5 ± 0.1 kg/cm2 Blowing	CP 1 10.5	CP 2 10.5	CP 1	CP 2
	Time Taken to fall BP Pr. From run to full service	MR 9.5 TO 9.8 kg/cm2 closing 6 to 10 sec.	CP 1 9.5	CP 2 9.5 kg	CP 1	CP 2
	b) Setting of CP Safety Valves.	11.5 ± 0.35 kg/cm2	CP 1	CP 2	CP 1	CP 2
			11.5	11.5	11.5	
	c)Setting of safety valve (Aux. CP.)	MR 8.5 ± 0.25 kg/cm2	8.5 kg/cm2		8.5 kg/cm2	
22	Check BP,BC,FP,MR,RS and AFI gauges Working to check	OK/ Not Ok	CAB-1	CAB-2	CAB-1	CAB-2
			OK	OK	OK	OK
			OK	OK	OK	OK

23	Check Working of parking brake by BPPB & Manually (wheel No. 2, 6, 7, 11)	Pr. In Gauge 6.0 + 0.15 kg/cm2 Apply pr. 0 kg/cm2 Release- Pr. 6.0 kg/cm2	CAB-1	CAB-2	CAB-1	CAB-2
			OK	OK	—	—
24	Check Operation /Leakage Emergency exhaust valve Panel Mounted	Ok/ Not Ok	check OK		check OK	
25	Check leakage of Duplex Check Valve	Ok/ Not Ok	—		—	
26	Check all air brake pipe line leakage. Brake cylinder leakage. Panel leakage.	Ok/ Not Ok	check OK			
27	BP/FP angle cock operation and leakage. to check , Tightness to check Hose pipe condition to check.	Ok/ Not Ok	CAB-1	CAB-2	CAB-1	CAB-2
	condition of Bogie Isolating cock	Ok/ Not Ok	CAB-1	CAB-2	CAB-1	CAB-2
28	Check BP/FP Rest Plate on both side cattle guard .Dummy Of FP/BP PIPE	Ok/ Not Ok	CAB-1	CAB-2	CAB-1	CAB-2
			OK	OK	OK	OK
29	Check Stenciling on BP/FP angle Cock & color Code. (BP-Green, FP- White)	Ok/ Not Ok	CAB-1	CAB-2	CAB-1	CAB-2
			OK	OK	OK	OK
	Check BP FP hose pipe details.	Make & Mfd. date to note	CAB-1		CAB-1	
			FP- BONY- 02-24		FP-SIDHARTH 11/25	
			BP- MANDEEP- 03-25		BP- MRE 06/25	
			CAB-2		CAB-2	
			FP- SCIPR- 02-25		FP- FEB/2025	
30	Check Working of PTDC in Knorr Bremse (KBIL)	Ok/ Not Ok	CAB-1	CAB-2	CAB-1	CAB-2
			—	—	—	—
31	Check F-70 control card LED stable condition to check	Ok/ Not Ok	CAB-1	CAB-2	CAB-1	CAB-2
			OK	OK	OK	OK
32	Check Cock No.136,70,74 IN open and 47 in close position to check.(F-70)	Ok/ Not Ok	CAB-1	CAB-2	CAB-1	CAB-2
			OK	OK	OK	OK

	Check Electrical junction box with Relay Control Portion (RCP) Multi-Purpose Input/output Node (MPIO) Power Supply Junction Box (PSJB) Check Electronic Vigilance Computer (EVC) cables & connectors (KBIL)	Check tightness of connectors to ensure proper locking and Check Operation & Visually check.	check ok	—
	Check Electronic Rack(E-70) Mother Board, PCU, LPO Card, Control card, Blender Card & VCU.	Check tightness of connectors to ensure proper locking and Check Operation & Visually check.	check ok	checked ok
34	Check Auxiliary manifold Isolating Cocks (TC1, TC2, FC, UL, CB, PAN1, PAN2, SAND1, SAND2, Feed Pipe cock, SIFA cock, Kaba Key) (KBIL)	Check for correct orientation of Isolating Cocks handles	check ok	—
35	Check Auxiliary manifold Magnet Valves (UL, 21, PAN1, PAN2, SAND1 & SAND2)	Ensure that connectors of Magnet valve are fixed Properly.	check ok	ok
36	A)Auxiliary manifold	Check the working and condition of valve. Repair/Replace, if any problem found.		
	Feed Valve		check ok	—
	Over flow Valve		—	—
	Check Valve		—	—
	Pressure Reducing Valve		—	—
	Impulse magnet valve Parking Brake		—	—
	Pressure Governors		—	—
	Safety Valves (AR-SV and AC-SV)		—	—
	Panto. pressure gauge		—	—
	B)EPCU functions Items		—	—
	Control Portion (16-CP, 20-CP, ER-CP, BP- CP,BC-CP) and Distributor Valve (KBIL)		—	—
	Pneumatic Equalizing Reservoir Actuator (PER COS) (KBIL.)		—	—
	Brake Controller Cab1 and Cab2 (KBIL.)		—	—
	Emergency Valve, NB11, PTDC(Driver brake valve), PRV, check valves, unloader valve, oil separator, Auto drain valves, Microphone horns, horn operating valves, sanding valve, 821 vent valve (KBIL.)		—	—
	Check Valve (CP-NRV) AND Pressure Governors Switch (KBIL.)		—	—
37	CHECKING PARAMETER			
	Compressor oil sample lab testing			
	Compressor lubes oil pressure record.			
	Check compressor oil level and leakage of oil	check ok	check ok	
	Check for any abnormal sound	check ok	check ok	ok
	Check for tightness of Assembly and air leakage from pipe connection of aux. Compressor	check ok	check ok	ok
	Conduct air brake self-test from the both control stands.	—	—	—

PRESSURE SWITCH SETTING :

Sr No.	Name of Pressrue Switch	Standard Value kg/cm2		IRC		FRC	
		Falling Pr.	Rising Pr.	Falling Pr.	Rising Pr.	Falling Pr.	Rising Pr.
1	Pantograph	4.5	5.5	—	—	—	—
2	Emergency brake	4.20	3.0	4.2	3.0	4.2	3.0
3	Auxiliary comp	7.0	8.0	7.0	8.0	7.0	8.0
4	Brake F P	5.0	5.6	5.0	5.6	5.0	5.6
5	Vigilance controll	1.5	2.0	1.5	2.0	1.5	2.0
6	Flow indicator	4.0	5.0	4.0	5.0	4.0	5.0
7	Bc bogie pr. Switch	0.2	0.65	0.2	0.65	0.2	0.65
8	Main compressor 1	8.0	10.0	8.0	10.0	8.0	10.0
9	Main compressor 2	8.0	10.0	8.0	10.0	8.0	10.0
10	Low MR pr	5.6	6.4	5.6	6.4	5.6	6.4
11	Direct brake pr. switch	0.2	0.65	0.2	0.65	0.2	0.65
12	QPDJ	3.3	3.6	3.3	3.6	3.3	3.6

SCHEDULE ACTIVITY

1)	CCB I schedule	Action Taken	Attend By
	All CCB valve remove (16-CP, 20-CP, ER-CP, BP-CP, BC-CP), Distributor Valve, PER COC .821 VENT VALVE, Feed valve, over flow measuring valve	—	—
	Both EBV to remove, PTDC (driver brake valve) to remove <i>AS 545</i>	Remove	Rakesh
	Below Item Cable Connectors tightness Check. a) Electrical junction box with b) Relay Control Portion (RCP) c) Multi-Purpose Input/output Node(MPIO) d) Power Supply Junction Box (PSJB) e) JUNCTION BOX ASSEMBLY f) Electronic Vigilance Computer (EVC) cables & connectors (KBIL)	cleaning	Satish
	CCB 2.0: LT switch resistance to check (standard limit: 0.04Ω - 2.0 Ω)	—	—
2)	CCB II SCH		
	All CCB valve to fit (16-CP, 20-CP, ER-CP, BP-CP, BC-CP), Distributor Valve, .821 vent valve, FEED VALVE, OVER FLOW MEASURING VALVE	—	—
	Both EBV to fit, PTDC (driver brake valve) TO FIT	oil fitting	Satish
3)	SANDING SCHEDULE		
A)	Check Sand box, cover gasket, Ejector and replace gasket if necessary.	changed	Satish vishesh
B)	Check all sander pipes for alignment and check fasteners also.	fitting	
C)	Check and ensure tightness of Sand box foundation bolt.	torqueing	
D)	Ensure sander pipe height 60mm above rail level.	checked	
E)	Sand to fill in all sand boxes.	filled	
F)	Sand box foundation bolts removed check length of bolt Refit and torque with lock tight.	torqueing	

4) WIPERS AND WASHER SCHEDULE		
A)	Remove the wiper motor assembly, Water Pump and Water tank.	changed ok
B)	Replace the wind screen wiper blades. Replacement should be done on condition basis.	changed ok
C)	Check the wiper and parallel arm assembly for wear and damage. Replace the arm or any part if there is excessive wear or damage.	changed ok
D)	Wiper -switch to be overhauled. Change all rubber components	changed ok
E)	FITMENT the wiper motor assembly, Water Pump and Water tank.	changed ok
F)	Wiper assembly looseness scoring and cracks. Renew any defective components. CHANGE ISOLATING COCK 3/8	changed ok
G)	Check the water level in reservoir no 1 and 2 and top up water	changed ok
5) FP BP ANGLE COCK SCH.		
A)	All angle cocks must be overhauled thoroughly duly changing all the rubber items	changed ok
B)	Check provision of BP/FP protection plates for welding, if damaged same should be attended.	changed ok
C)	Check Coal Dust of Trap Chamber. Clean and couple	checked ok
D)	Both side FP And BP angel cock & hose pipes change	changed ok
6) UNDER FRAME BRAKE VALVE (As per OEM's manual)		
A)	Remove all under frame brake valve (NB-11 valve, MR safety valve, Both unloader valve and NRV, Both auto drain valve (with drip cup), Both emergency exhaust valve, Both RS valve.)	changed ok
B)	Fitment of brake valve (NB-11 valve, MR safety valve, Both unloader valve and NRV, Both auto drain valve (with drip cup), Both emergency exhaust valve, Both RS valve.)	changed ok
7) UPPER FRAME BRAKE VALVE		
A)	Remove All Brake Valve (MR Safety Valve, RS Emergency Valve, Horn Valve And Horn Body, Sanding Valve, ARSV, ACSV, Filter (AC, MREP Filter, MR Filter, MR, 20 P Filter, MR2 Assembly), Per Cos, Filter Element (BP, Auxr, DBP Filter), Check Valve.)	Removed ok
B)	Fitment All Brake Valve (MR Safety Valve, RS Emergency Valve, Horn Valve And Horn Body, Sanding Valve, ARSV, ACSV, Filter (AR, MREP Filter, MR Filter, MR, 20 P Filter, MR2 Assembly), PER Cos, Filter Element (BP, Auxr, DBP Filter), Check Valve.)	fitted ok
8) AUXILLAIRY MANIFOLD SCHEDULE		
(A) REMOVE OF : Isolating cocks (TC1, TC2, FC, UL, CB, PAN1, PAN2, SAND1, SAND2, FEED pipe cock, SIFA cock, Kaba Key) Isolating cock 1 1/4, vented (AIRDRYER, MR) Isolating cock without vent 1/4 inch (MREP, DBP, BC) DRAIN COCK RESERVOIR 1/2 INCH Isolating cock 3/8 OF HORN , Pressure Reducing Valve, ALL PRESSURE Governor, Magnet Valve (UL, 21, PAN1, PAN2, SAND1, SAND2), Check Valve.		changed ok
(B) FITMENT OF : Isolating cocks (TC1, TC2, FC, UL, CB, PAN1, PAN2, SAND, SAND2, FEED pipe cock, SIFA cock, Kaba Key) Isolating cock 1 1/4, vented (AIRDRYER, MR) Isolating cock without vent 1/4 inch (MREP, DBP, BC) DRAIN COCK RESERVOIR 1/2 INCH Isolating cock 3/8 OF HORN , Pressure Reducing Valve, ALL PRESSURE Governor, Magnet Valve (UL, 21, PAN1, PAN2, SAND1, SAND2), Check Valve.		changed ok
9) GAUGES RACK SCHEDULE		
A)	Remove and piping to check and replace.	changed ok

B)	Fit		
10)	AIR DRYER(As per MP.ML18)		
	Air Dryer To Remove	Removed ok	Vinay
	Fitment Of Air Dryer And Isolating Cock 1 ¼ Vented	fitted ok	
11)	AUXILIARY COMPRESSOR		
	Aux. Compressor to remove	Removed ok	Shah
	Aux. Compressor to fit and oil to fill		
	Replace the delivery hose between the auxiliary compressor and brake frame.	changed ok	
	Clean the CPA externally and check tightness of connections.	checked ok	Sush
12)	MAIN COMPRESSOR (SD Vol.F13)		
	Both Compressor to remove and send to section	Removed ok	
	Conduct RDPT of compressor & motor foundation bracket(BY LAB)	checked ok	
	Both compressor to fit and ensure tightness of rope wire and alignment of compressor	checked ok	
	Soft and hard mount AVM to change	changed ok	Orionot Ved
	Check tightness of Nut bolts of all Guards of CP. Check Fan.	checked ok	
	Replace CP delivery pipe	changed ok	
	Check the impeller housing for any external hits and breakages	checked ok	
	Replace copper interconnecting pipe b/W I/C AND A/C	checked ok	
13)	Brake Pneumatics :(M+R, Vol. D3, SD Vol. F11, F12, 13, 14)		
	Clean & Test the all pressure switches as part of the brake unit, check settings. Replace any faulty or defective switches. Check intactness of mounting bases	checked ok	Vinay
14)	Both bogie brake cylinder overhaul		
	All 08 nos brake cylinder overhaul to do		
	All bogie brake cylinder flexible pipe line to change	changed ok	Sush

JE /SSE PNEUMATIC SECTION NAME AND SIGN: Pulay

JE /SSE COMPRESSOR SECTION NAME AND SIGN: P. Gauram

SSE INCHARGE NAME AND SIGN: [Signature]

03/02/2026

QR/UF/F/02
WHEEL CHANGING INFORMATION

To

Laboratory (QAG-LAB)

Record Room (QAG-MI)

Date :- 25-3-26

Following Axle Nos. fitted on the loco

32461

IOH

Sr. No.	Axle No.	Type of Axle	Remark
01	8319	NBC	All wheel RD Branded. Read Vahni Date - 05-03-26
02	5401	"	" " "
03	4324	"	" " "
04	7438	FAG	" " DHD Date - 07-03-26
05	4409	NBC	" " "
06	3637	"	" " "

T.T. Done on date : 22-3-26 (Due to Dig less)

Wheel dia. CE : 1090.33 HE : 1090.89-mm

Wt CE - Date 05-03-26 } JS-Sinha
HE - Date 16-03-26 }



JE/SSE(M)UF

QAG-UF